

Laboratory Project 1: Electromyogram Circuit

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Abstract-You will build an electromyogram (EMG) circuit consisting of electrodes, pre-amps, and a differential amplifier. After designing, constructing, and testing the circuit, you will connect electrodes to your biceps muscle and measure the small voltages generated by the activity of neurons terminating on muscle fibers. By recording these signals on an oscilloscope, you will create EMGs. You will analyze the EMGs to determine the rate of neural activity associated with lifting various weights.

I. PREPARATION

For Lab 1, which will last about three weeks, you will need the parts listed in Table I. You may purchase these parts from the stockroom next to the lab or purchase them elsewhere.

TABLE I
PARTS LIST FOR LAB 1B

Item	Qty	Description
1	1	Breadboard
2	1	Breadboard Wire Kit
3	2	Red LED
4	1	Green LED
5	1	LM324 Quad Operational Amplifier
6	4	Resistors (values determined during lab)
7	1	10 Ω Resistor
8	3	1 k Ω Resistor
9	1	1 M Ω Resistor
10	1	0.1 μ F Capacitor
11	3	Electrodes

II. LEARNING OBJECTIVES

- 1) Learn about voltage dividers and understand why pre-amps are necessary to increase current drive of weak signals from electrodes.
- 2) Learn how to derive equations for op-amp circuits, such as pre-amps and the differential amplifier, using Kirchhoff's and Ohm's laws.
- 3) Understand how to design a differential amplifier to meet practical constraints.
- 4) Determine the relationship between neuromuscular activity and force applied by a muscle by measuring electromyograms.

III. INTRODUCTION

A. Overview

In Lab 1 you will build an amplifier circuit to measure electromyograms (EMG's) and view them on an oscilloscope. The EMG is a plot showing the tiny voltages produced by your biceps muscle and picked up by electrodes, as shown in Fig. 1. You will attach the electrodes to your arm, and they will feed into pre-amp circuits that output the same voltage as is present on the electrode but with a higher current-drive capability. You will explore why this is necessary in the first part of this lab. After building and testing the pre-amps, you will add a differential amplifier to complete EMG circuit. The differential amplifier will You will measure EMG's while lifting different amounts of weight and complete a simple study of how weight affects the measured EMG signal.

B. Electromyograms

EMG studies in general are useful for assessing the health of the neuromuscular system, since certain diseases, such as multiple sclerosis, slow down or even suppress normal nerve and muscle firing. In addition, several research groups have recently studied the possibility of using EMG signals to control artificial limbs for patients who have lost an extremity; the EMG signal would be obtained from a surviving portion of the limb and would represent the patient's central nervous system's desire to move the limb in a certain direction with a certain force.

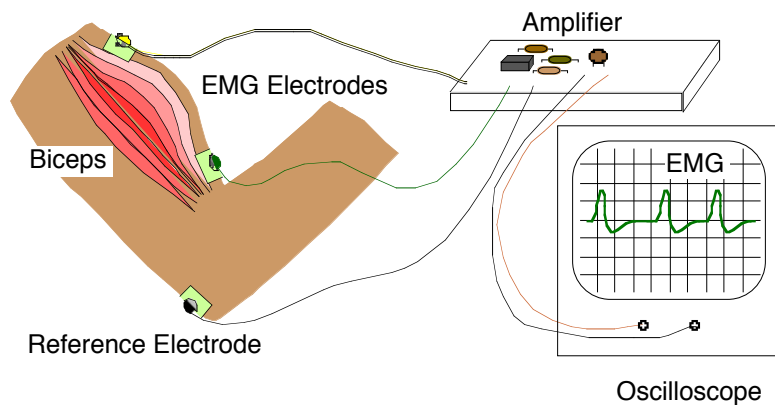


Figure 1. System for measuring an electromyogram.

C. Op-Amps

The Lab 1 circuit uses an op-amp (short for operational amplifier) integrated circuit (IC or “chip”). The op-amp IC comes in a package with 8 pins. For Lab 1, we may treat the op-amp as a basic circuit component whose function is to measure a voltage drop at its inputs and output a voltage that is about 100,000 (or 100k) times the input voltage drop. Fig. 2, below, illustrates the op-amp as it appears in a circuit diagram and how it is connected to the pins on the package. Note that the op-amp IC actually contains four op-amps, allowing us to build four amplifier circuits with one chip. Although we will omit the details of the circuitry inside the op-amp because it is beyond the scope of the course, you may find this information by searching for "[On Semi LM324 data sheet](#)" on the web.

Just as cell phones and mp3 players require batteries, an op-amp requires power supplies. We connect +15V and -15V to the pins labeled V_{CC} (pin 4) and V_{EE} (pin 11), respectively. The myDAQ supplies these voltages, allowing us to avoid wasting batteries. When you use the device to measure your EMG, however, it is safer to use batteries for +9V and -9V power.

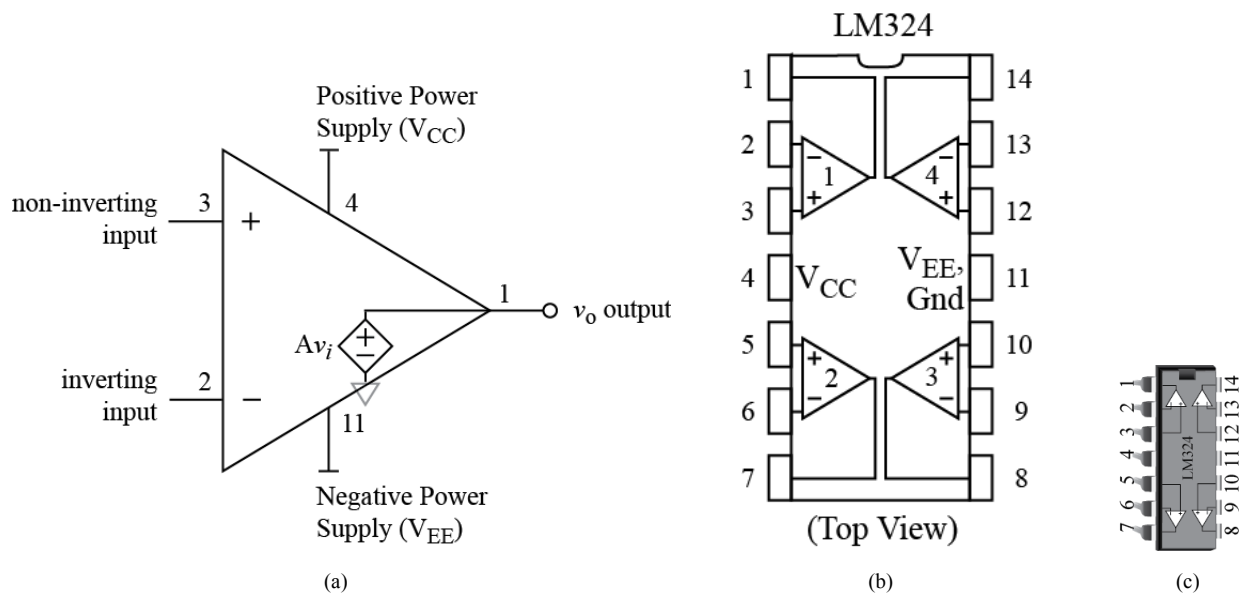


Figure 2. LM324 operational amplifier: (a) model, (b) pins on package, (c) picture.

D. Design Project Overview

You will complete the design and construction of an EMG circuit that receives input from two neighboring electrodes placed on the biceps of the upper arm plus a reference electrode placed on the elbow. The objective of the circuit is to measure the small voltage drop between the electrodes on the biceps caused by neuromuscular activity. It turns out that a differential amplifier is useful for amplifying small voltage drops.

The power in the signals picked up by the electrodes, however, is minute. Attaching the electrodes directly to a differential amplifier would draw too much current from the electrodes, causing their voltage to drop to almost zero. Consequently, we use pre-amps to create higher-power signals. The pre-amps can output higher current at the same voltage as the electrodes while drawing virtually zero current. The outputs of the pre-amps can then drive a differential amplifier.

Fig. 3 shows a block diagram of the EMG circuit with the pre-amps and differential amplifier that you will build in this lab. You will connect the output voltage, v_3 , to an oscilloscope to record an EMG.

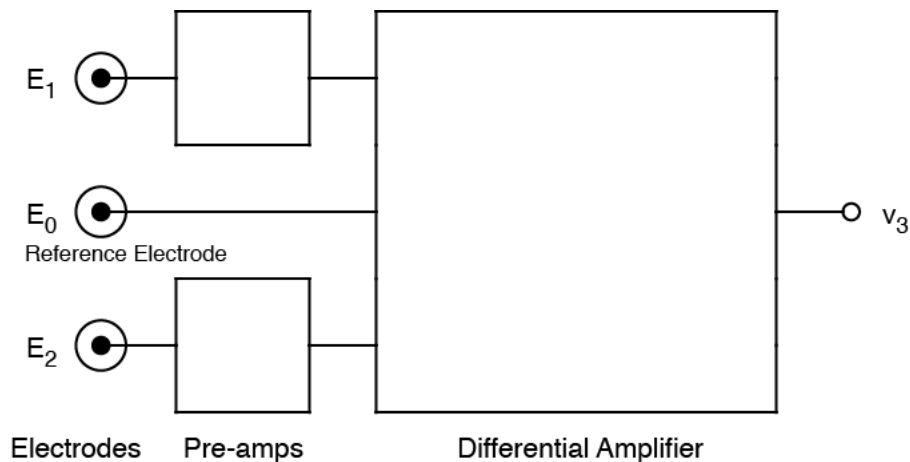


Figure 3. Block diagram of electromyogram circuit.

IV. MODELING ELECTRODES AND PRE-AMPS

A. Rationale

In this part of the lab, you will demonstrate that using the electrodes to directly drive the differential amplifier would result in signals too small to be accurately measured. This scenario is similar to considering what would happen if we tried to use a 12V camera battery instead of a car battery to start a car—the smaller battery would be unable to supply enough current to the starter motor, and output voltage of the small battery would drop to almost 0 V.

B. Electrode Model

Each electrode is a metal pad with conductive gel that contacts the skin to pick up voltages on the order of 90 mV, (only a fraction of which is detected by the finite-sized electrode), generated by neuromuscular activity. The voltage sources are inside muscles, well below the skin, and so are separated from the skin by the resistance of the muscle, other tissues, and skin. The resistance of the skin, being dry on the surface, is quite high. An approximate model of the resistance is a 1 M Ω resistor. Thus, we may model an electrode (and muscle) as a voltage source on the order of 90 mV in series with a 1 M Ω resistor, as shown in Fig. 4.

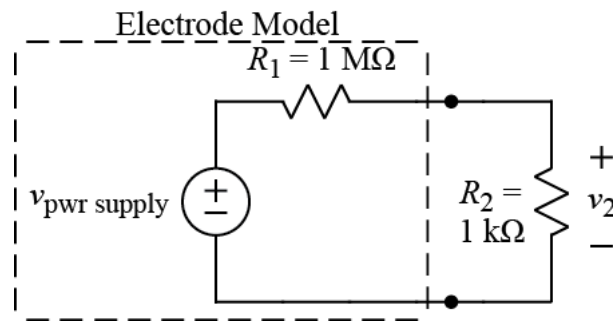


Figure 4. Circuit model of an electrode driving differential amplifier: schematic.

C. Model of Electrode Driving Differential Amplifier

It would be convenient if we could connect electrodes directly to a differential amplifier that would magnify the voltage drop between electrodes. We now explore what would happen if we tried to do this.

If we consider how much input current the differential amplifier we will be using requires, we may model the differential amplifier input as approximately a 1 k Ω resistor. This value is obtained by dividing the voltage at the input of the differential amplifier by the current that flows into the differential amplifier. In other words, we use Ohm's law. Attaching this resistor to the electrode model yields the circuit, shown in Fig. 4, that we will use to simulate the electrode driving the differential amplifier directly. You will show that the electrodes would be unable to drive the differential amplifier.

To mimic the circuit in Fig. 4, build the circuit shown in Fig. 5 or use your own circuit layout. (Fig. 5 also includes +15V and -15V supplies and power indicator LED's that show when the +15V and -15V power is on. Since they allow current to flow in only one direction, try the LED's in both directions if they fail to light.) In place of an electrode, use a resistor and the myDAQ function generator as a variable power supply voltage. The function generator is found on the same menu of instruments as the multimeter and oscilloscope. Normally, it is used to output a time-varying waveform like a sinusoid or square wave, but we will use it to produce a DC voltage. To do that, set the amplitude knob to 0 V_{pp} and set the DC Offset knob to the desired output voltage. You may ignore the frequency knob since the amplitude of the time-

varying part of the waveform is zero. The function generator output is on the myDAQ analog output line AO0. Note that the AGND output from the myDAQ must also be connected in order for the function generator to work. There are two outputs for AGND that are connected together inside the myDAQ.

Record the measured voltage drop across R_2 in the second column of Table II. When you have completed the second column of Table II, use the voltage-divider formula from class to fill in the third column of Table II. The input voltage from the myDAQ function generator is " $v_{\text{pwr supply}}$ " in Fig. 4. Be sure to record what you are doing, including the voltage-divider formula, in your notebook.

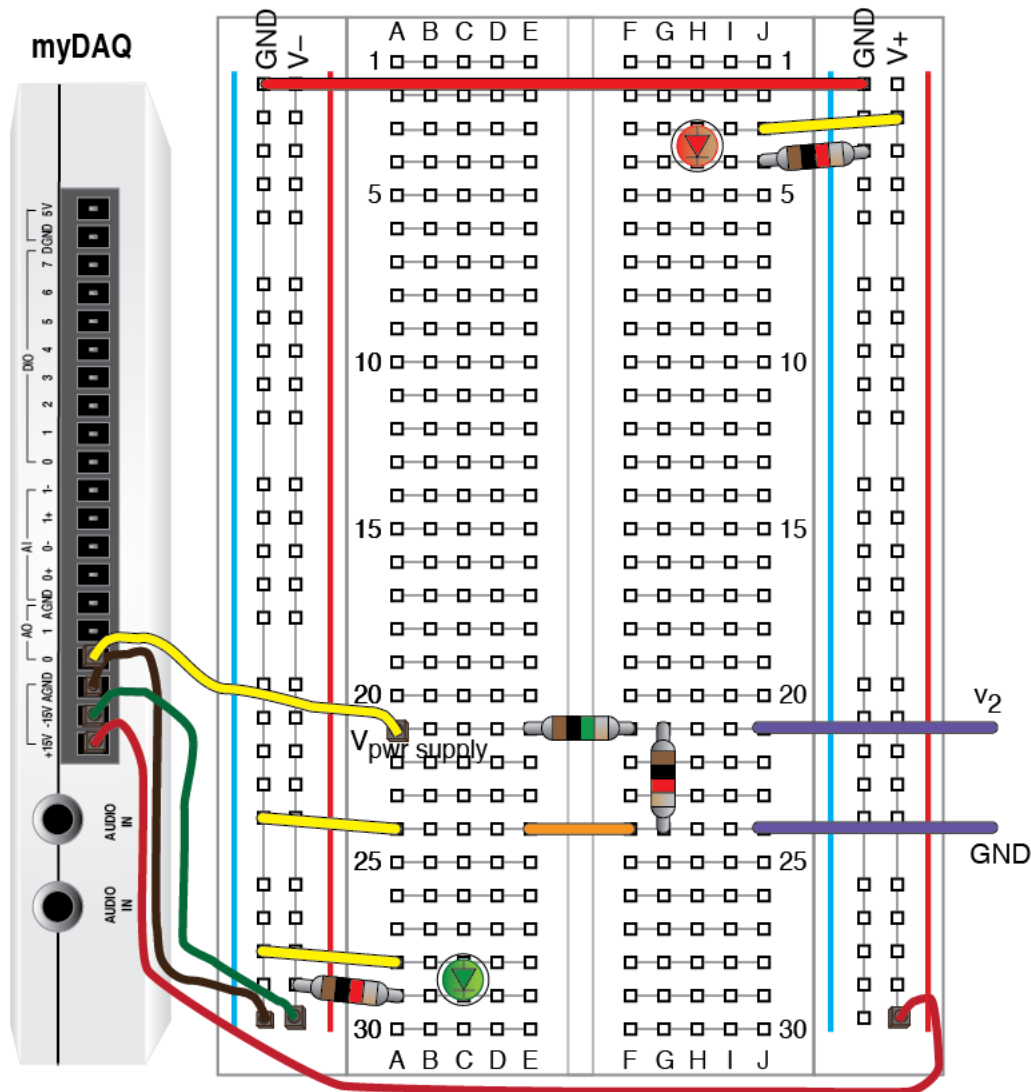


Figure 5. Circuit model of an electrode driving differential amplifier: breadboard layout.

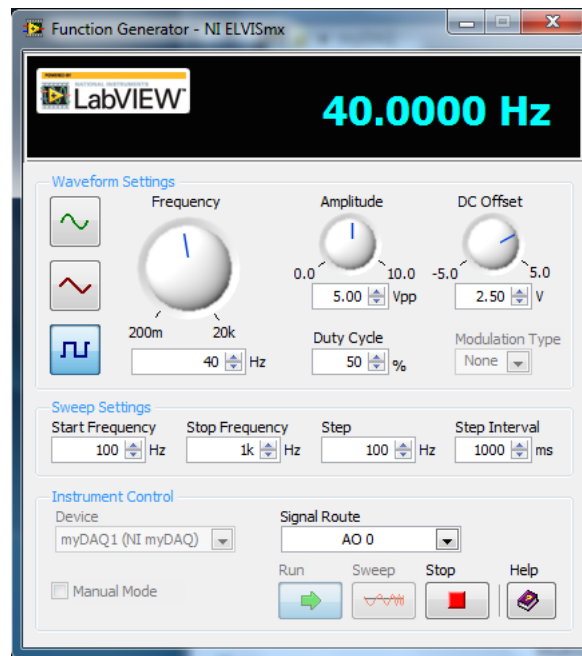


Figure 6. The myDAQ function generator.

TABLE II
MODEL OF ELECTRODE DRIVING DIFFERENTIAL AMPLIFIER DIRECTLY

Power Supply Voltage setting (V) (myDAQ DC offset)	v_2 Voltage (V) (calculated)		v_{pwr} supply Voltage (V) (measured)	v_2 Voltage (V) (measured)
0 V				
1 V				
2 V				

D. Model of Pre-Amp Driving Differential Amplifier

When we add a pre-amp to the electrodes, we get an output voltage that is the same as the electrode voltage but the output current can be much higher. If we model the entire electrode and pre-amp circuit as a Thevenin equivalent, the Thevenin resistance (or output resistance) is about $10\ \Omega$. To see how the pre-amp improves the strength of our electrode signal, repeat the process in Section C using the circuit shown in Fig. 7. Note that only R_1 changes. Record your data in Table III. As before, R_2 simulates the input of the differential amplifier.

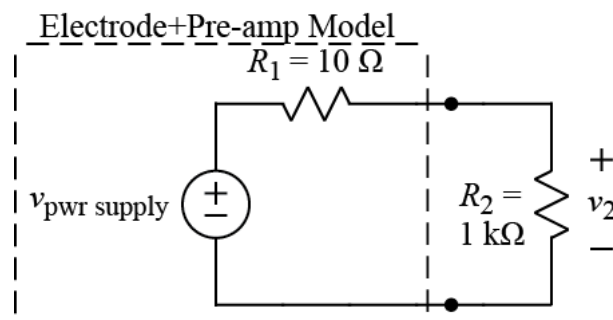


Figure 7. Circuit model of electrode + pre-amp driving differential amplifier.

TABLE III
MODEL OF ELECTRODE+PRE-AMP DRIVING DIFFERENTIAL AMPLIFIER

Power Supply Voltage setting (V) (myDAQ DC offset)	v_2 Voltage (V) (calculated)		$v_{\text{pwr supply}}$ Voltage (V) (measured)	v_2 Voltage (V) (measured)
0 V				
1 V				
2 V				

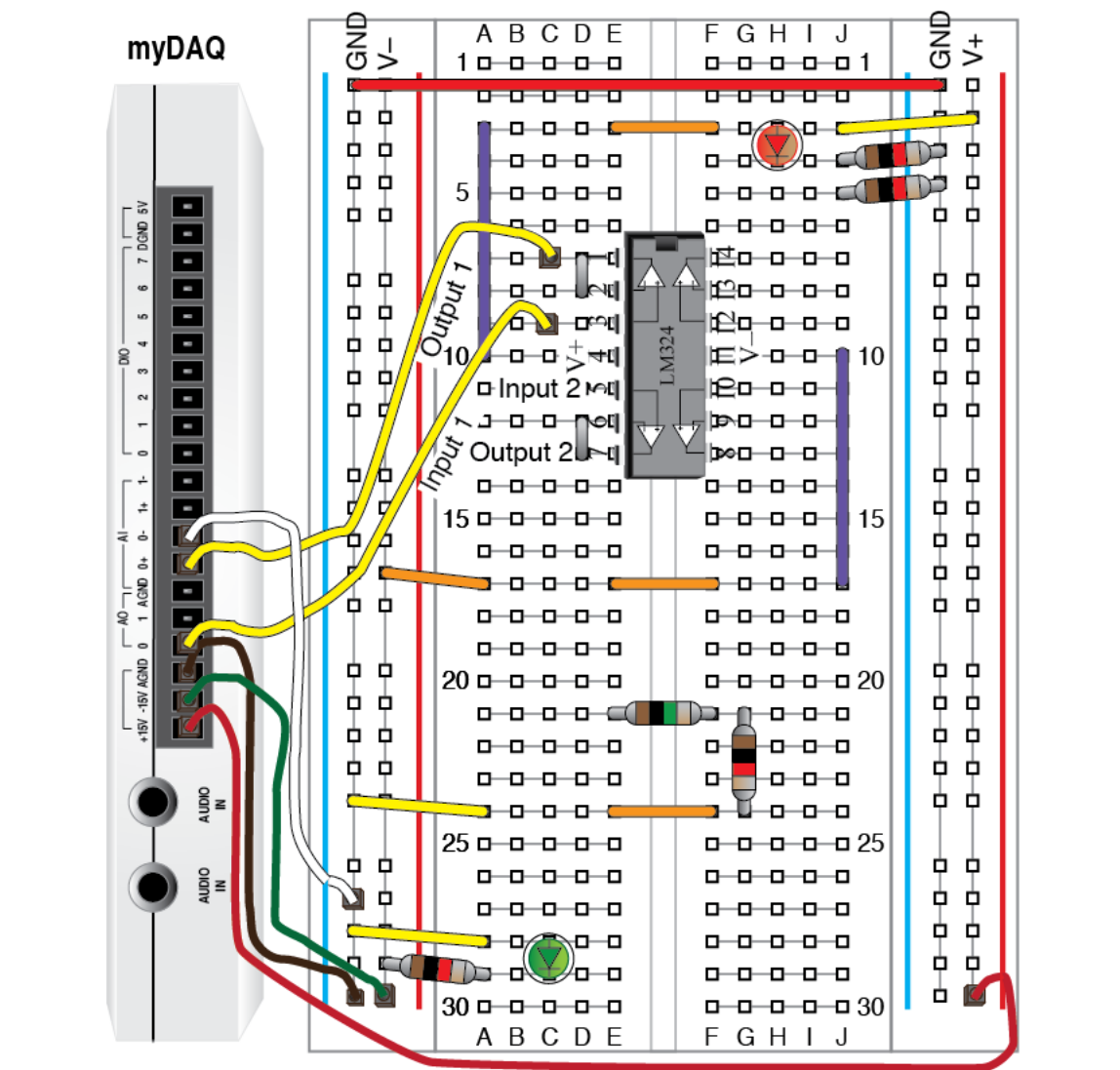
When you have filled out Tables II and III, determine which circuit gives an output that is closer to the value of the input voltage from the power supply. Comment in your lab notebook on the results and explain why the pre-amps are needed in the EMG circuit. Note that, when used for an actual EMG, the input signal to the circuit will only be a few millivolts instead of the higher values used here. If the signals get much smaller, they sink below the measurement noise.

V. CONSTRUCTION AND TESTING OF PRE-AMPS

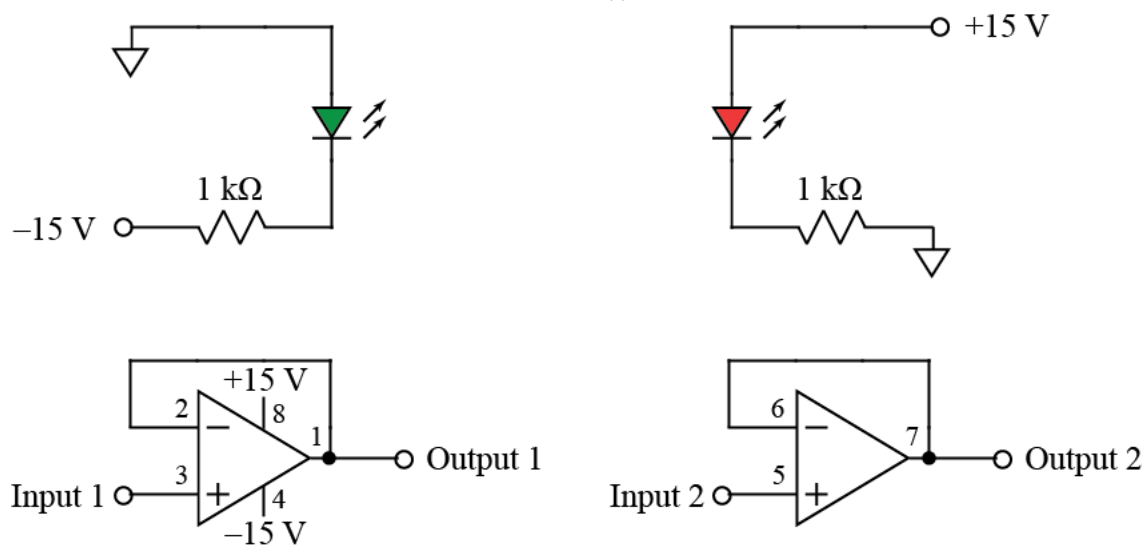
A. Construction

Using the diagram in Fig. 8a, construct the two pre-amp circuits whose schematic diagram is shown in Fig. 8b. The LM324 is an op-amp IC that you may purchase in the stockroom (or elsewhere). Note that all points in the circuit schematic marked with a triangle, meaning they are at reference (i.e., 0V or what is often referred to as "ground"), are connected together. Note also, that there are two very short vertical wires near the op-amp chip. And the voltage divider from earlier is left on the board for use later on.

To test the op-amp pre-amp circuit, you will use a function generator and an oscilloscope. Fig. 9 shows the myDAQ oscilloscope. After making the connections shown in Fig. 8, set up the function generator to output a 1V sinusoid with zero DC offset at a frequency of 1 kHz. Press the "Auto Scale" button the oscilloscope to see the waveform coming out of the pre-amp. This signal is on the AI0+ input of the myDAQ. This wire is labeled "Output 1" in Fig. 8. You may move this wire around to measure different signals. Use it to measure the signal going into "Input 1" of the circuit, and use it to measure the signal coming out at "Output 1". Record the waveforms in your notebook. Make a careful drawing of the oscilloscope screen, showing the scale for the horizontal axis (time) and the vertical axis (voltage). The time and voltage represented by one box is printed on oscilloscope screen. Be careful to include the value of each in your drawing. If all goes well, you will have seen two sinusoids that are the same size.



(a)



(b)

Figure 8. Pre-amp circuits: (a) breadboard layout, (b) schematic diagram.

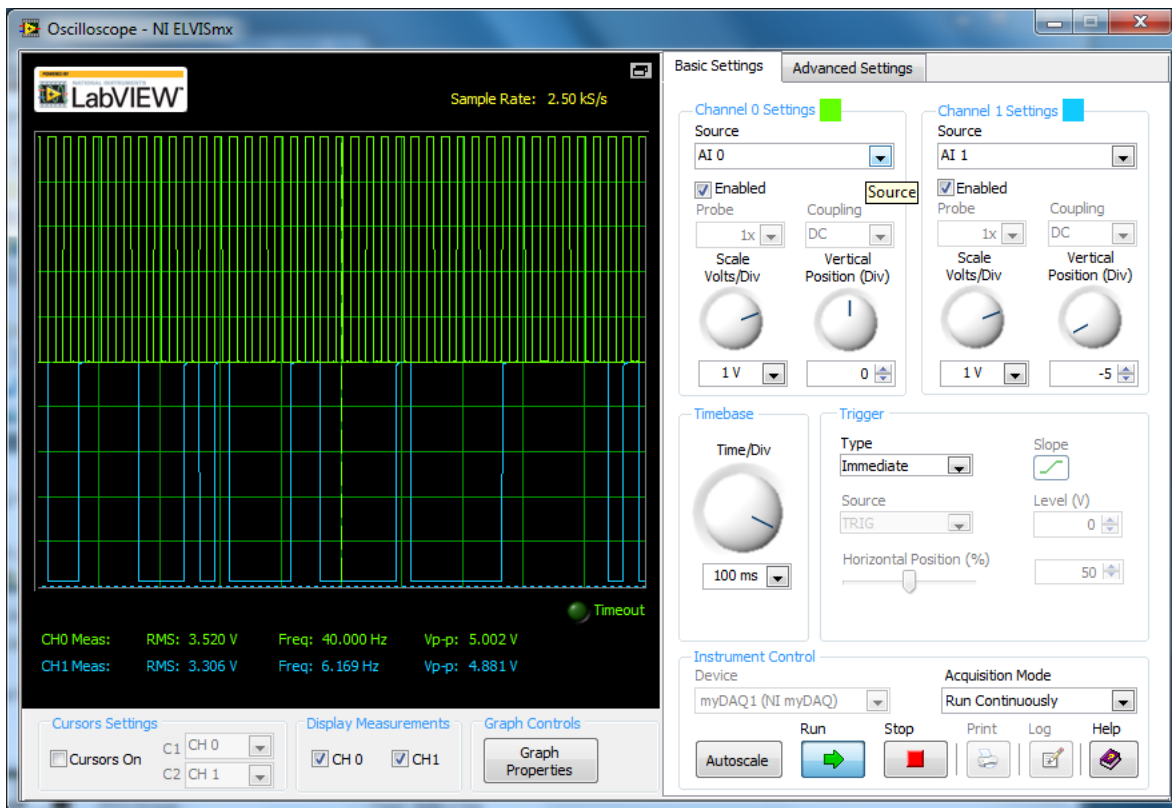


Figure 9. myDAQ oscilloscope.

VI. DERIVING AN EXPRESSION FOR THE DIFFERENTIAL AMPLIFIER OUTPUT

In this part of the lab, you will use the voltage-divider formula to derive an expression for the output voltage of the differential amplifier circuit shown in Fig. 10. Note that the power supplies for the op-amp are omitted from the schematic, but +15 V must be connected to pin 4 and -15V must be connected to pin 11 of the op-amp chip. Also, the wire to the reference electrode must be connected to the reference in your circuit, which is connected to the AGND output from the myDAQ. (The reference is also where the ground leads for oscilloscope probes and the function generator are connected.)

A. Deriving the Expression for v_3

When the output of an op-amp is connected back to the "-" input via a resistor, the op-amp output effectively adjusts to make the voltage drop across the + and - inputs zero. This is referred to as negative feedback. It is possible to show that this occurs because the op-amp output voltage is a large multiple of the voltage drop across the + and - inputs. We omit this analysis here, owing to its relative complexity, and we just assume that the voltage drop across the + and - inputs is zero. Using this assumption, we can find an expression for the output voltage of the op-amp, (i.e., v_3), as a function of electrode voltages, v_1 and v_2 , by the following procedure:

- 1) Use the circuit model in Fig. 11(a) and the voltage-divider formula to find voltage drop v_+ across R_4 .
- 2) Use the circuit model in Fig. 11(b) and Kirchhoff's and Ohm's law to find voltage drop v_- across R_2 and v_3 . In other words, v_- is the sum of the voltage drops across R_2 and v_3 .
- 3) Set $v_+ = v_-$ and solve for v_3 in terms of v_1 and v_2 .

Armed with the expression for v_3 , we can now show that the differential amplifier may be designed to amplify only the difference in voltages v_1 and v_2 , thus reducing noise in the EMG measurements.

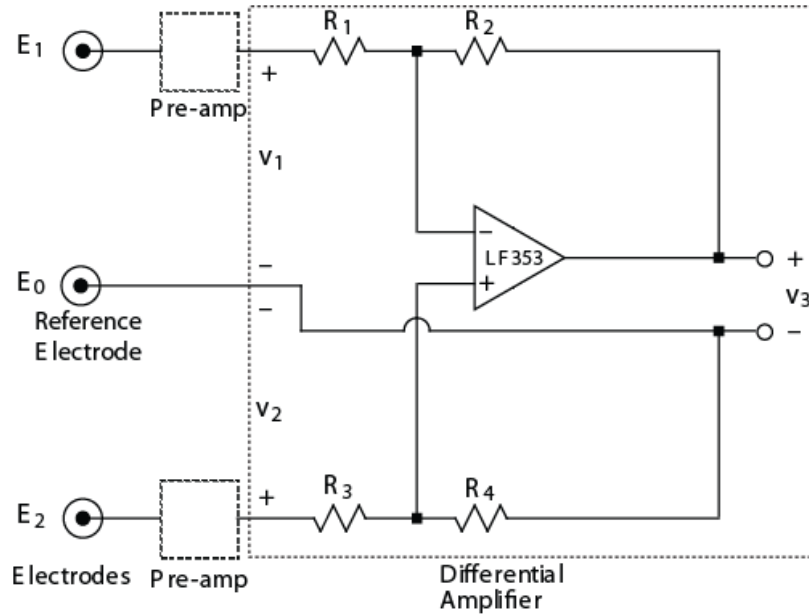


Figure 10. Schematic diagram of differential amplifier.

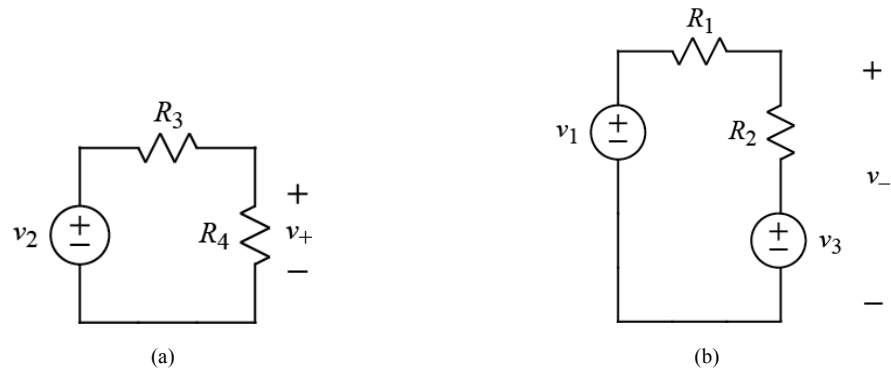


Figure 11. Sub circuits for analyzing differential amplifier: (a) v_+ value, (b) v_- value.

B. Differential Gain

We want v_3 to be proportional to only the difference (v_{dm} for "differential mode voltage") between v_1 and v_2 . This is helpful because any offset voltage (v_{cm} for "common mode voltage") common to both electrodes that is caused by a source other than nerve and muscle activity will be cancelled out. (Furthermore, electronic noise generated by non-ideal characteristics in our circuit will typically be the same for both electrodes and will be cancelled out.) The differential and common-mode voltages are defined as follows:

$$v_{cm} \equiv \frac{(v_2 + v_1)}{2} \quad (1)$$

$$v_{dm} \equiv v_2 - v_1 \quad (2)$$

To determine how to make the common-mode response (or gain) zero, rewrite the formula for v_3 in terms of the common-mode signal, v_{cm} , and the differential-mode signal, v_{dm} , defined in (1) and (2). Begin this process by making the following substitutions:

$$v_1 = v_{cm} - \frac{v_{dm}}{2} \quad (3)$$

$$v_2 = v_{cm} + \frac{v_{dm}}{2} \quad (4)$$

Now show that v_3 is a function of only v_{dm} if the ratio of R_1 to R_2 is the same as the ratio of R_3 to R_4 . To do so, first rewrite v_3 in terms of the following ratio, $\mathfrak{R}$:

$$\mathfrak{R} = \frac{R_1}{R_2} = \frac{R_3}{R_4} \quad (5)$$

(Note that it may be helpful to consider the reciprocals of terms or to divide the top and bottom of terms by a convenient resistor value [leaving the term's value unchanged], in order to express things in terms of $\mathfrak{R}$.) Then show that v_{cm} disappears from the expression for v_3 . Now that you have derived the expression for v_3 , you are ready to design the differential amplifier. That is, you are ready to find the values of R_1 through R_4 . Having found those values, you will build the circuit.

VII. DESIGNING, BUILDING, AND TESTING THE DIFFERENTIAL AMPLIFIER

In this part of the lab, you will complete the following tasks:

- 1) Determine the resistor values to use in the differential amplifier for a gain of 500.
- 2) Build and test the differential amplifier.
- 3) Measure the gain of the differential amplifier

A. Resistor values for a gain of 500

Keeping the preceding analysis in mind, design a differential amplifier for the electromyogram circuit that meets the following four design objectives:

- 1) The differential gain of the circuit is to be 500. The differential gain is the term multiplying v_{dm} in the equation for v_3 derived above. This makes the output as large as possible without causing the output to "saturate" by reaching the op-amp power supply voltages. (The output is limited by the power-supply voltages, resulting in clipping distortion of the output waveform if the voltage reaches the level of the power supply.)
- 2) The common mode gain of the circuit is to be zero. (See Equation (5) in previous section.)
- 3) The input resistances, R_1 and R_3 , are to be the same for both inputs. This will help cancel out less than ideal output characteristics of the pre-amps that might be amplified by any asymmetry in the differential amplifier's inputs. (Explanation of this is beyond the scope of this course.)
- 4) The input resistances, R_1 and R_3 , must be high enough that the input current never exceeds the maximum current, (20 mA), that the op-amp in the pre-amp can supply. Use the worst-case voltage drop that can possibly occur across any resistor, which is $+15\text{ V} - -15\text{ V} = 30\text{ V}$ across R_1 or R_3 to determine the minimum R allowed. (Note that actual voltages out of our pre-amps will be small, and exact values are unknown.)
- 5) The maximum resistor values used in your circuit (R_1 , R_2 , R_3 , and R_4) should be limited to about 1 M Ω . This is because even a small noise current in our circuit can create a

significant voltage across a high-valued resistor. Thus, we limit the resistor size in order to limit the voltage that small noise-currents will create.

The resistor values satisfying the above requirements are not unique, so your answer may vary a bit from what other people decide to do. List the values of R_1 , R_2 , R_3 , and R_4 prominently in your lab notebook.

B. Building and Testing the Differential Amplifier

Your next task is to build and test the differential amplifier. Because the circuit has a high gain, the circuit must be tested with small input voltages. Fig. 12 shows how a voltage divider can be used to create a small input voltage from a power supply voltage of several volts. The resulting input to the pre-amps mimics what the electrode signal looks like. If two power supplies were available, it would be possible to have a second voltage divider to mimic Electrode 2 and test both inputs at once. Since only one extra supply, (the function generator), is available, however, you will test your circuit using a voltage-divider on only one input at a time. The other input will be connected to reference.

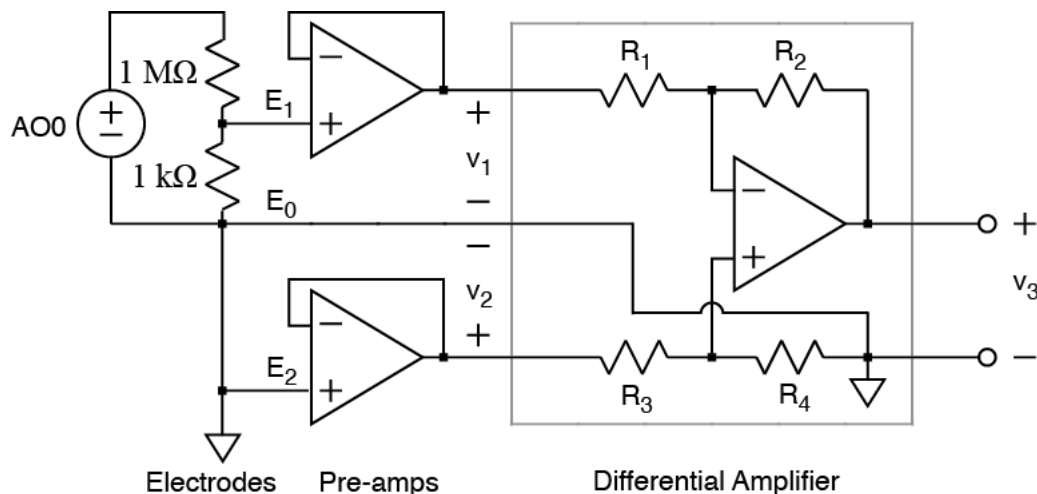


Figure 12. Testing differential amplifier using voltage divider.

Fig. 13 shows one way to layout the voltage divider on your breadboard for testing the EMG circuit. (You will probably want to use your own layout, however.) Measure the differential-amplifier output voltage for input voltages, $AOO = 0\text{ V}$, 1 V , and 2 V from the function generator, for input to the voltage divider whose output is $E1$. Then move the voltage divider to the $E2$ input, connect the $E1$ input to reference, and repeat the process for input 2. In your notebook, fill in copies of Tables IV and V, below.

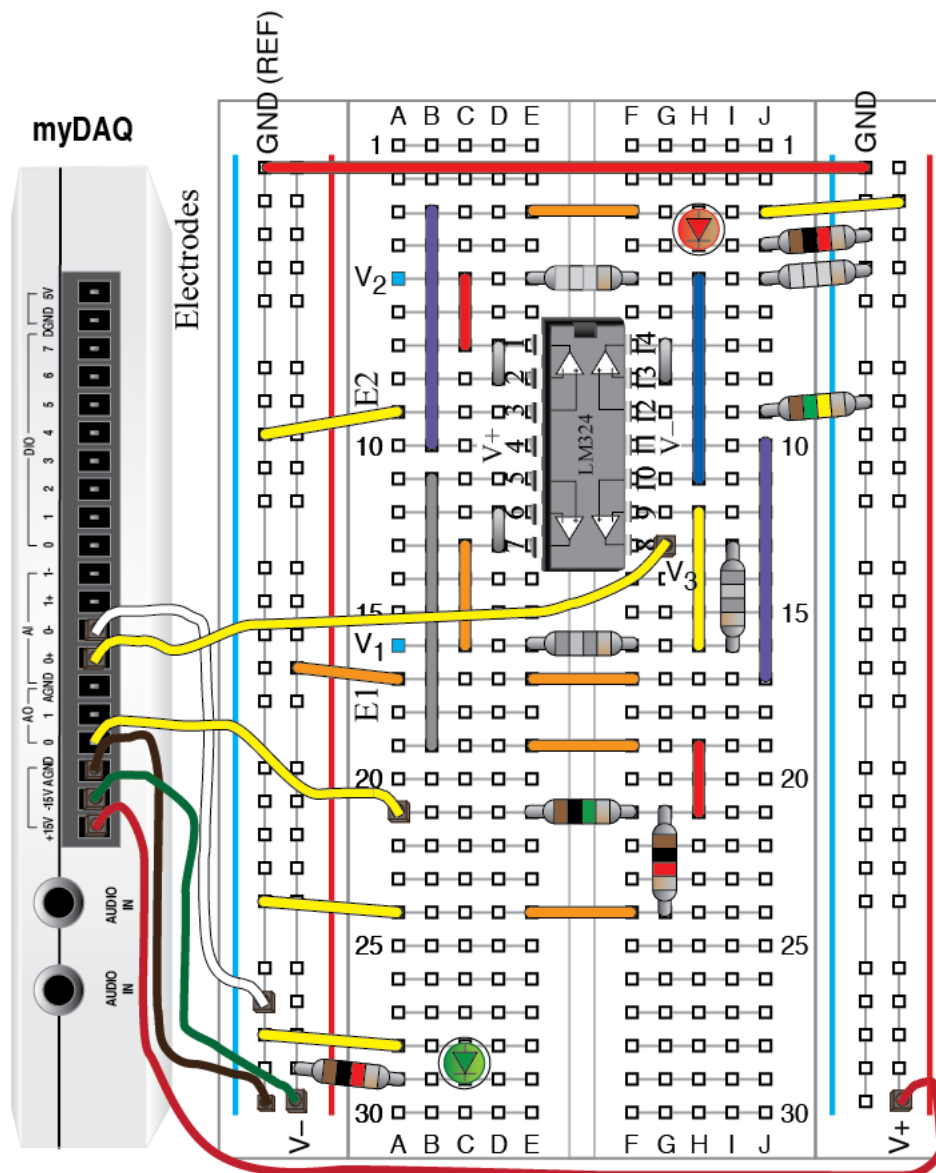


Figure 13. Circuit for testing differential amplifier using voltage divider.

TABLE IV
MEASUREMENTS OF DIFFERENTIAL AMPLIFIER WITH v_1 INPUT

Input V Voltage setting (V) (myDAQ DC offset)	v_1 Voltage (V) (calculated)	v_2 Voltage (V) (calculated)	$v_2 - v_1$ Voltage (V) (calculated)	v_3 Voltage (V) (measured)
0 V		0 V		
1 V		0 V		
2 V		0 V		

TABLE V
MEASUREMENTS OF DIFFERENTIAL AMPLIFIER WITH v_2 INPUT

Input V Voltage setting (V) (myDAQ DC offset)	v_1 Voltage (V) (calculated)	v_2 Voltage (V) (calculated)	$v_2 - v_1$ Voltage (V) (calculated)	v_3 Voltage (V) (measured)
0 V	0 V			
1 V	0 V			
2 V	0 V			

C. Measuring the Gain of the Differential Amplifier

Verify that the gain of the differential amplifier is close to 500. Warning: calculating the gain requires at least two measurements from which the slope of a line may be determined. To calculate the gain, use your data in Tables IV and V to make a plot of v_3 versus $v_2 - v_1$. Using the `polyfit` function in MATLAB, draw a straight line through the data and find the slope of the line. The slope is the gain of the circuit.

Note that your plot does not pass through the origin. Our method of calculating the gain eliminates a large constant offset in the output that results from a non-ideal offset voltage across the + and – inputs of the op-amp in the differential amplifier. That is, any real op-amp is slightly flawed. This offset voltage is only a few millivolts and represents the voltage across the + and – inputs that the op-amp interprets as exactly zero volts. In many applications this offset voltage may be neglected. In the differential amplifier circuit, however, the offset voltage is similar in size to the input signals and also gets multiplied by 500, causing a significant output voltage even when the two signals driving the differential amplifier are zero. A capacitor is used later on to eliminate this offset when measuring the EMG.

VIII. MEASURING AND ANALYZING EMG'S

A. Building and Testing the EMG Circuit

The complete EMG circuit is shown in Fig. 14 but with resistor values missing. Draw a schematic diagram of the complete EMG circuit in your notebook. Include the circuit for the fourth op-amp (high-pass filter) on your schematic. What does this circuit for the unused op-amp do? (Ask your TA.)

Using the layout in Fig. 15 or one of your own design, add the fourth op-amp (high-pass filter) to your circuit. Use $C = 1 \mu\text{F}$ and $R_5 = 1 \text{ M}\Omega$. Procure electrodes (available in the stockroom) and check out snap wires from the equipment window to connect the electrodes to your breadboard.

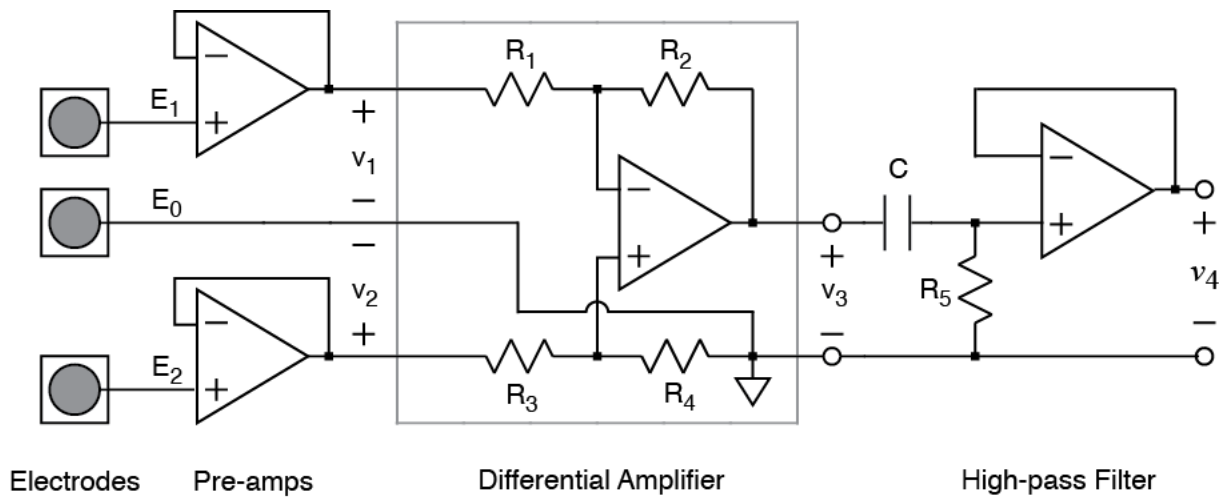


Figure 14. Complete EMG circuit schematic.

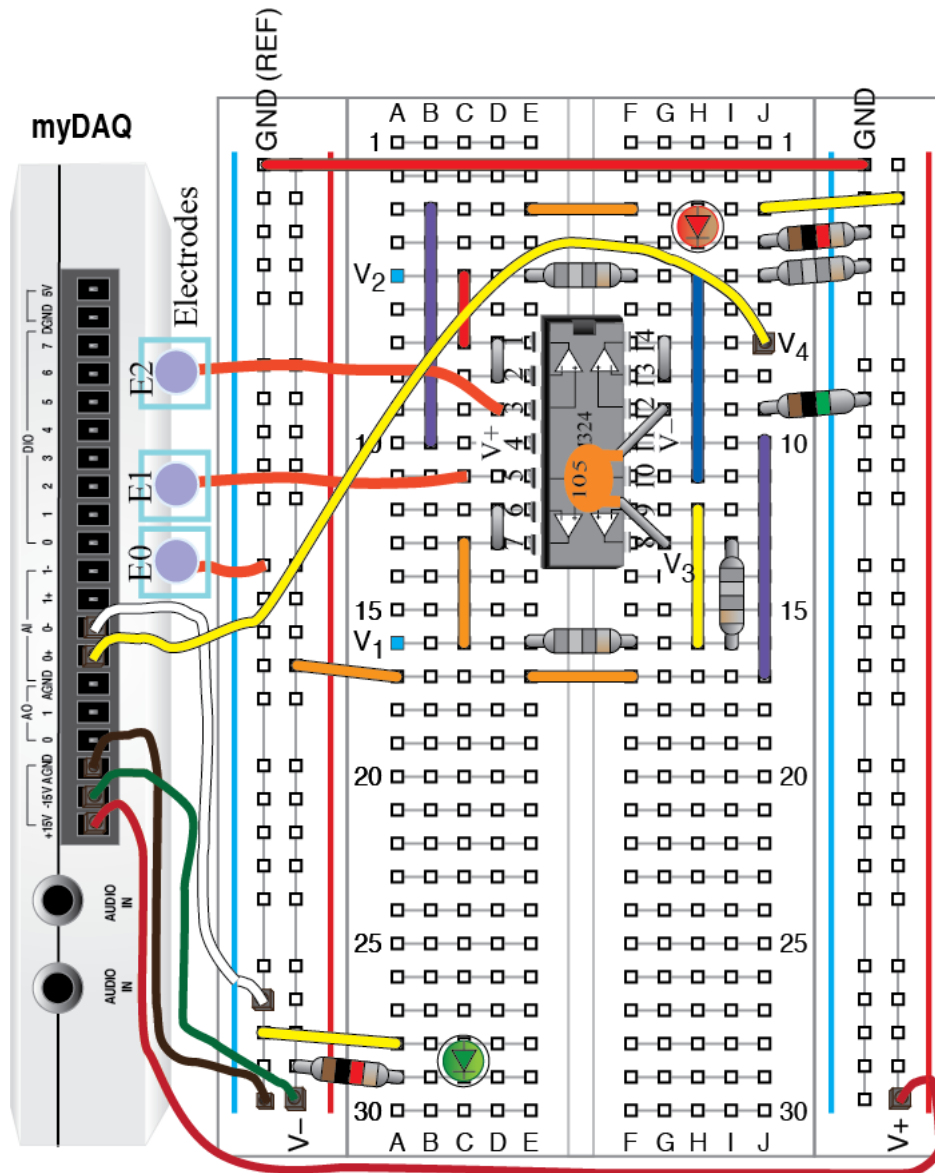


Figure 15. Complete EMG circuit.

B. Measuring EMG's

In this part of the lab, you will complete the following tasks:

- 1) Connect electrodes to your biceps and measure EMG's while you are lifting various weights.
- 2) Plot and comment on the power in your EMG signal as a function of the weight lifted.

As a safety precaution, use two 9 V batteries as the power supplies for your electromyogram circuit when measuring EMG's with actual electrodes. That is, replace the +15V and -15V power supplies from the myDAQ with two 9V batteries.

Connect electrodes to your biceps—the muscle on the top of the upper arm that bulges when showing off your strength. Place two electrodes, measuring the voltages, E_1 and E_2 , going into the preamps, about three inches apart on the upper and lower end of the biceps slightly toward the outside of the muscle. Place the third, reference electrode, on the elbow. (Avoid placing the reference electrode on muscle.)

Connect the output of the electromyogram circuit to the myDAQ oscilloscope. To eliminate the large constant vertical (DC) offset of the waveform, the fourth op-amp acts as a high-pass filter between the differential amplifier output and the oscilloscope probe. The capacitor in the circuit acts to block DC.

Observe the waveform on the oscilloscope and capture an example of the waveform that you can plot in MATLAB on a computer. (Use the "log" button on the oscilloscope to access the data recorded on the oscilloscope screen.) Print out copies of the waveform for both the lab notebook and report. Note that you will need to record data for several different waveforms below, too.

C. Power versus Weight for EMG signals

Write MATLAB code to calculate the average "power" of the recorded waveform by calculating the average value of voltage squared:

$$p = \frac{1}{N} \sum_{i=1}^N (v_{3i} - \bar{v}_3)^2 \quad (6)$$

where

p is the average "power" of the EMG circuit output signal

N is the number of sample points

v_{3i} is the i th sample of the EMG circuit output voltage

$\bar{v}_3 = \frac{1}{N} \sum_{i=1}^N v_{3i}$ is the average value (or DC offset, or vertical offset) of the recorded waveform

(Note that p actually has units of voltage squared rather than power, but p is equal to the power we would have if we connected a 1Ω resistor to the output of the circuit. Also, we subtract the average voltage of the waveform because our signal, given the output filter, should have zero offset. Any offset present in the final signal is a result of undesired DC offsets in the circuit.)

D. Plot of Electromyogram Power versus Weight

Record waveforms and use your MATLAB code to determine the average EMG signal output power, p , while holding the lower arm horizontal with no weight, one unit of weight, two units of weight, and three units of weight. Choose weights such that the three-unit weight requires significant but comfortable effort when held with the lower arm horizontal. When performing the tests with weights, keep your joints in a constant position as much as possible.

Using MATLAB, make a plot of p versus weight. Comment on the shape of this plot.

NOTEBOOK AND REPORT

Turn in a copy of your laboratory notebook pages and a separate formal report. Refer to the grading information on the course website for the section numbering to use while writing the formal report. Use the IEEE format for typesetting. Information about the IEEE format, including a template file, is available on the course website. Additional information about writing the report and keeping a notebook is on the course website. Note that MATLAB plots, if they are listed in the contents of the report, must appear both in the laboratory notebook *and* the formal report.

A point breakdown for grading of the lab notebook and the report are on the course website. As you write your report, imagine that you are writing a self-contained document for your supervisor at work. Assume that the supervisor does not have access to your notebook, and make sure that what you write is understandable to someone with a technical level comparable to your own. For example, you might write the report as though it is instructions to next year's students describing how to do the lab. If the report is well written, you may wish to show it to potential employers as an example of your work.

ACKNOWLEDGMENT

K. Furse assisted in the writing of this handout.